

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Tech. (ME/EE/CL) Examination

May 2013

ME 101 Engineering Graphics

Date: 11.05.2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Figures drawn in the question paper are not to the scale.
6. Dimensions shown in the figure are in mm.
7. Show all the necessary dimensions in the solution.

SECTION - I

Q.1 (a) Choose appropriate answer by rewriting the question

[05]

- 1 The R.F. of the scale on a mini-drafter is
(a) 0 (b) 1 (c) 10 (d) none of these
- 2 Name the curve which has zero eccentricity.
(a) Ellipse (b) Parabola (c) Hyperbola (d) Circle
- 3 A point whose front view and top view are above XY is situated in
(a) First angle (b) Second angle (c) Third angle (d) Fourth angle
- 4 If top view of a line is a point, its front view is
(a) parallel to XY and of true length
(b) parallel to XY and of apparent length
(c) perpendicular to XY and of true length
(d) perpendicular to XY and of apparent length
- 5 A pyramid is cut by a plane parallel to its base removing apex, the remaining part is known as
(a) truncated (b) frustum (c) sectioned (d) prism

Q. 1(b)i) Give the application of AutoCAD.

[02]

ii) Construct a scale of 1 cm = 1 m to read metres and decimeters and long enough to measure up to 14 metres. Show on this a distance equal to 12.4 m.

[05]

iii) A point P is 40 mm away from H.P. & 30 mm away from V.P. Draw its projections in all possible positions.

[03]

OR

Q.1 (b) The foci of an ellipse are 120 mm apart and the minor axis is 70 mm long. Draw the ellipse by concentric circle method. Also draw the normal and tangent at any point on the curve.

[10]

Q. 2 A line MS, 80 mm long is inclined at 45° to H.P. and 30° to the V.P., its end M is in the H.P. and 40 mm in front of V.P. Draw the projections.

[08]

OR

Q. 2 A 100 mm long line PQ has its end P 10 mm above the H.P. and 70 mm in front of the V.P. The line is inclined at 60° to the H.P. and 30° to the V.P. Draw its projections.

[08]

- Q.3 Fig. 1 shows pictorial view of an object. Draw
(1) Front view in the direction 'X', (2) Top view.

[12]

SECTION - II

- Q.4 Draw the projections of a regular hexagon of 25 mm side having one of its sides in the H.P. and inclined at 60° to the V.P. and its surface making an angle of 45° with the H.P. [07]

OR

- Q.4 PQRS is a rhombus of diagonal PR is 60 mm and QS is 40 mm the corner P is in the HP and the plane is inclined to the HP such that the plan appear is square. The plan of diagonal PR makes an angle of 20° to the VP. [07]
- Q.5 (a) A square pyramid of 40 mm base side and 65 mm long axis has its base on the H.P. and all the edges of the base equally inclined to the V.P. It is cut by a section plane perpendicular to the V.P. and inclined at 45° to H.P. and bisecting the axis. Draw the sectional top view and true shape of the section. [09]
- (b) A pentagonal prism, side of base 35 mm and height 70 mm, is resting on HP on its base with one of the edge of the base inclined at 60° to the VP. It is cut by an AIP inclined to HP by 30° passing through a point on the axis 15 mm from top end of the axis. Draw the development of the cut prism. [09]

OR

- Q.5 (a) A hexagonal pyramid of 30 mm base side and 60 mm long axis has an edge of its base on the H.P. and the axis inclined at 30° to the H.P. The edge of the base on which it rests is inclined at 45° to the V.P. Draw its projections. [09]
- (b) A square pyramid with 30 mm side of base and 50 mm long axis is resting on its base such that all the sides of the base are equally inclined to the V.P. It is cut by a section plane perpendicular to V.P. and inclined at 60° to H.P. The section plane passing through the mid-point of the axis. Draw the development of the cut pyramid. [09]
- Q.6 Fig. 2 shows the orthographic projections of an object in first angle projection system. Draw its isometric view. [10]

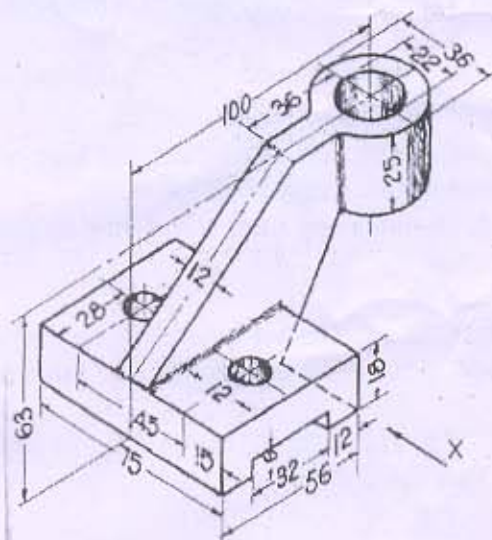


Figure 1

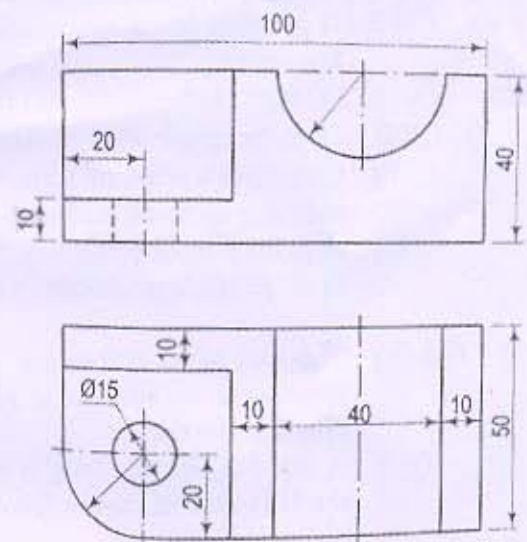


Figure 2
